

Ex-1

Pk6667
RTY667
2

```
1 #include<iostream>
2 using namespace std;
3 class Car{
4     string CarNumber;
5     public:
6     static int totalCars;
7     Car(string CarNumber){
8         this -> CarNumber =CarNumber;
9         totalCars++;
10    }
11    void display(){
12        cout<<CarNumber<<endl;
13    }
14    static void CarCounter(){
15        cout<<totalCars;
16    }
17 };
18 int Car::totalCars=0;
19 int main(){
20     Car C1("Pk6667");
21     Car C2("RTY667");
22     C1.display();
23     C2.display();
24
25     Car::CarCounter();
26
27 }
```

Ex-2

```
1 #include<iostream>
2 using namespace std;
3 class BankAccount{
4     string AccNumber;
5     public:
6     static float interest_rate;
7     BankAccount(string AccNumber){
8         this -> AccNumber = AccNumber;
9     }
10    void Display(){
11        cout<<AccNumber<<endl;
12    }
13    static void ShowInterestRate(){
14        cout<<interest_rate;
15    }
16 };
17 float BankAccount::interest_rate = 3.5;
18 int main(){
19     BankAccount B1("Yash15262");
20     BankAccount B2("Yash67889");
21     B1.Display();
22     B2.Display();
23     BankAccount::ShowInterestRate();
24 }
```

Yash15262
Yash67889
3.5

Ex-3

```

1  #include<iostream>
2
3
4  using namespace std;
5  class Book{
6      public:
7          const string ISBN;
8          Book(string isbn):ISBN(isbn){}
9
10     void Display() const{
11         cout<<ISBN<<endl;
12     }
13 };
14 int main(){
15     Book B1("Taare-Zameen-24536");
16     Book B2("Harry-Puttar-4567");
17
18     B1.Display();
19     B2.Display();
20 }

```

```

Taare-Zameen-24536
Harry-Puttar-4567

```

Ex-4

```

1  #include<iostream>
2  using namespace std;
3  class circle{
4      public:
5          float radius;
6          circle(float r):radius(r){}
7
8          float Area()const{
9              return 3.14*radius*radius;
10         }
11 };
12 int main(){
13     circle c1(2.5);
14
15
16     cout<<"Area: "<<c1.Area();
17
18 }

```

```

Area: 19.625

```

Ex-5

```

1  #include<iostream>
2  using namespace std;
3
4  class Cpu{
5
6      string generation;
7      public:
8      Cpu(string g){
9          generation = g;
10     }
11     void DisplayInfo(){
12         cout<<"CPU Gen: "<<generation<<endl;
13     }
14
15 };
16
17 class Computer{
18     string Model;
19     string Brand;
20     Cpu CPU;
21     public:
22     Computer(string model,string brand,string Gen):Model(model),Brand(brand),CPU(Gen){}
23
24     void DisplayInfo(){
25         cout<<"Model: "<<Model<<endl;
26         cout<<"Brand: "<<Brand<<endl;
27         CPU.DisplayInfo();
28     }
29 };
30 int main(){
31     Computer c1("Victus","HP","i7-13th");
32     c1.DisplayInfo();
33
34 }

```

```
Model: Victus
Brand: HP
CPU Gen: i7-13th
```

Ex-6

```
1 #include<iostream>
2 using namespace std;
3 class employee{
4     int id;
5     float salary;
6     public:
7     void SetDetails(int Id,float Salary){
8         salary = Salary;
9         id =Id;
10    }
11    void Display(){
12        cout<<id<<endl<<salary<<endl;
13    }
14 };
15 int main(){
16     employee e[5];
17     for(int i=0;i<5;i++){
18         e[i].SetDetails(i+1,37000.76);
19     }
20     for(int i=0;i<5;i++){
21         e[i].Display();
22     }
23 }
```

```
1
37000.8
2
37000.8
3
37000.8
4
37000.8
5
37000.8
```

Task-1

```
1 #include<iostream>
2 using namespace std;
3 class LibraryMember{
4     string name;
5     static int totalmember;
6     public:
7     LibraryMember(string n){
8         name = n;
9         totalmember++;
10    }
11    void displayName(){
12        cout<<"Name: "<<name<<endl;
13    }
14    static void displaytotalmember(){
15        cout<<totalmember;
16    }
17 };
18 int LibraryMember::totalmember = 0;
19 int main(){
20     LibraryMember m1("Yash");
21     LibraryMember m2("Harsh");
22     LibraryMember m3("Aisha");
23     LibraryMember m4("Usman");
24     m1.displayName();m2.displayName();m3.displayName();m4.displayName();
25     LibraryMember::displaytotalmember();
26
27
28
29 }
```

```
Name: Yash
Name: Harsh
Name: Aisha
Name: Usman
4
```

Task-2

```

1  #include<iostream>
2  using namespace std;
3  class schoolAccount{
4      string name;
5      int grade;
6      static float tutionfee;
7      public:
8      schoolAccount(string name,int grade){
9          this->name = name;
10         this->grade = grade;
11     }
12     void displayInfo(){
13         cout<<"Name: "<<name<<endl;
14         cout<<"Grade: "<<grade<<endl;
15     }
16     static void showfee(){
17         cout<<tutionfee;
18     }
19 };
20 float schoolAccount::tutionfee = 3567.5;
21 int main(){
22     schoolAccount a1("Yash",7);
23     schoolAccount a2("Harsh",6);
24     a1.displayInfo();a2.displayInfo();
25     schoolAccount::showfee();
26 }

```

```

Name: Yash
Grade: 7
Name: Harsh
Grade: 6
3567.5

```

Task-3

```

1  #include<iostream>
2  using namespace std;
3  class Vehicle{
4      const int vehicleID;
5      string modelName;
6      public:
7      Vehicle(string modelName,int ID):modelName(modelName),vehicleID(ID){}
8
9      void displayInfo()const{
10         cout<<"Model Name: "<<modelName<<endl;
11         cout<<"Vehicle ID: "<<vehicleID<<endl;
12     }
13 };
14 int main(){
15     Vehicle v1("Honda City",345);
16     v1.displayInfo();
17 }

```

```

Model Name: Honda City
Vehicle ID: 345

```

Task-4

```

1  #include<iostream>
2  using namespace std;
3  class Square{
4      float side_length;
5      public:
6      Square(float l){
7          side_length = l;
8      }
9      float CalculateArea() const{
10         return side_length*side_length;
11     }
12     void displayArea(){
13         cout<<"Area: "<<CalculateArea()<<endl;
14     }
15 };
16 int main(){
17     Square s1(4.5);
18     Square s2(2.0);
19     s1.displayArea();
20     s2.displayArea();
21 }

```

```

Area: 20.25
Area: 4

```

Task-5

```

1  #include<iostream>
2  using namespace std;
3  class SIM{
4      int strength;
5      public:
6      SIM(int s):strength(s){}
7      void ShowCarrier(){
8          cout<<"Strength: "<<strength<<endl;
9      }
10
11 };
12 class Phone{
13     string model;
14     SIM sim1;
15     public:
16     Phone(string model,int s):model(model),sim1(s){}
17     void display(){
18         cout<<"Model: "<<model<<endl;
19         sim1.ShowCarrier();
20     }
21 };
22 int main(){
23     Phone p1("Redmi Note 13",3);
24     Phone p2("Samsung s22",4);
25     p1.display();p2.display();
26 }

```

```

Model: Redmi Note 13
Strength: 3
Model: Samsung s22
Strength: 4

```

Task-6

```

1  #include<iostream>
2  using namespace std;
3  class Product{
4      string name;
5      int id;
6      float price;
7      public:
8
9      void Input(){
10         cout<<"Enter Name: ";
11         cin>>name;
12         cout<<"Enter ID: ";
13         cin>>id;
14         cout<<"Enter Price: ";
15         cin>>price;
16     }
17     void Display(){
18         cout<<"Name: "<<name<<endl;
19         cout<<"ID: "<<id<<endl;
20         cout<<"Price: "<<price<<endl;
21     }
22 };
23 int main(){
24     Product p[4];
25     for(int i=0;i<4;i++){
26         cout<<"---- Enter Product "<<i+1<<" Details----"<<endl;
27         p[i].Input();
28     }
29     for(int i=0;i<4;i++){
30         cout<<"--- Product "<<i+1<<" ---"<<endl;
31         p[i].Display();
32     }
33 }
34 }

```

```
---- Enter Product 1 Details----
Enter Name: Mellon
Enter ID: 88
Enter Price: 500
---- Enter Product 2 Details----
Enter Name: Pen
Enter ID: 44
Enter Price: 15
---- Enter Product 3 Details----
Enter Name: Water
Enter ID: 8
Enter Price: 32
---- Enter Product 4 Details----
Enter Name: Nugget
Enter ID: 55
Enter Price: 250
--- Product 1 ---
Name: Mellon
ID: 88
Price: 500
--- Product 2 ---
Name: Pen
ID: 44
Price: 15
--- Product 3 ---
Name: Water
ID: 8
Price: 32
--- Product 4 ---
Name: Nugget
ID: 55
Price: 250
```

Task-7

```
1  #include <iostream>
2  using namespace std;
3  class Patient
4  {
5      string name;
6      const int P_ID;
7
8  public:
9      Patient(string name, int id) : P_ID(id), name(name) {}
10
11     void display() const
12     {
13         cout << "Name: " << name << endl;
14         cout << "ID: " << P_ID << endl;
15     }
16 };
17 class Hospital
18 {
19     Patient *P[10];
20     static int PatientCount;
21
22 public:
23     void addNewPatient(string name, int id)
24     {
25         if (PatientCount < 10)
26         {
27             P[PatientCount] = new Patient(name, id);
28             PatientCount++;
29         }
30         else
31         {
32             cout << "Limit is Over" << endl;
33         }
34     }
```

Second Half:

```
34 }
35 void displayAll()
36 {
37     for (int i = 0; i < PatientCount; i++)
38     {
39         P[i]->display();
40     }
41 }
42 static void totalPatients()
43 {
44     cout << PatientCount << endl;
45 }
46 };
47 int Hospital::PatientCount = 0;
48 int main()
49 {
50     Hospital H1;
51     H1.addNewPatient("Yash", 998);
52     H1.addNewPatient("Mahboor", 977);
53     H1.addNewPatient("Qasim", 888);
54     H1.displayAll();
55     cout << "Total Patients: ";
56     Hospital::totalPatients();
57 }
```

```
Name: Yash
ID: 998
Name: Mahboor
ID: 977
Name: Qasim
ID: 888
Total Patients: 3
```